

Sharath Thomas

Senior DevOps Engineer

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PERSONAL PROFILE

Senior DevOps Engineer with 3+ years of experience building cloud-native platforms and CI/CD infrastructure for engineering simulation products. Proven track record of rapid prototyping, cross-team enablement, and turning ambiguous technical challenges into production-ready solutions across Azure, AWS, and Kubernetes environments. Published researcher with an MSc in Advanced Computer Science from the University of Leicester.

EXPERIENCE

Synopsys / Ansys Sheffield, UK

Senior Engineer Apr 2025 — Present

- Cross-team platform engineer and go-to contact for integration, deployment, and troubleshooting of Ansys cloud simulation platforms.
- Led the PoC for containerizing NVIDIA Omniverse Kit Apps (digital twin) and deploying them on OKAS and HPS.
- Built a custom UI for job submission and real-time streaming, enabling instant visual feedback for simulation users.
- Provisioned and managed Azure Kubernetes clusters (AKS) for scalable OKAS deployments.
- Contributed to an internal container registry platform with Terraform-provisioned infrastructure for automated image scanning, promotion, and delivery.

R&D Engineer II Dec 2024 — Apr 2025

- Designed and maintained GitHub Actions composite actions and reusable workflows for SAF SDK components and SAF-based applications.
- Automated checks, tests, builds, and release pipelines, reducing manual deployment efforts across multiple engineering teams.
- Drove the Ansys Fluent webapp streaming PoC, enabling browser-based simulation access via HPS and Ansys Cloud Burst Compute.
- Deployed and debugged AWS Parallel Clusters with Slurm queues for scalable HPS compute.
- Created Terraform modules for deploying platform components on single-node K3s clusters.

R&D Engineer Jul 2023 — Dec 2024

- Automated Ansys HPS deployment on VMs using Ansible, later migrating the Simulation Platform to a Docker Compose stack.
- Supported developers across multiple teams in deployment, maintenance, and complex troubleshooting.
- Built a PoC converting SAF platform components (portal API/UI, Keycloak, Nginx, OAuth2-Proxy) to Windows Services for customers unable to use Docker.
- Managed on-prem and cloud VMs across Azure and AWS for platform hosting and development.

DevOps Intern Jul 2022 — Jun 2023

- Created and maintained standardized CI/CD pipelines using GitHub Actions for solution applications.
- Migrated multiple repositories from an Azure DevOps monorepo to GitHub, preserving full git history and pipeline configurations.
- Built and containerized a React/Flask web application providing UI-based access to internal Ansible automation scripts.
- Packaged Python developer tools with Nuitka for streamlined internal distribution.

Samsung R&D Institute India Bengaluru, India

Nov 2019 — Aug 2020

- Developed a gesture recognition application for Samsung Watches using Tizen OS.
- Enabled smartwatch-based control of Samsung TVs through intuitive hand gestures for actions like channel switching and volume adjustment.
- Built sensor data processing pipelines for real-time gesture interpretation on wearable hardware.

ISA-VIT (International Society of Automation) Vellore, India

Dec 2017 — Jun 2020

- Progressed from Core Member to Technical Head of the student chapter, leading and mentoring 100+ members.
- Conducted weekly "Technitudes" sessions introducing students to emerging technologies in automation, IoT, and software development.
- Co-ordinated an international research conference for academic paper publications as part of the board.
- Organised hands-on workshops and events that made the chapter a go-to community for skill development on campus.

EDGATE Technologies Bangalore, India

Jun 2019 — Jul 2019

- Embedded Systems and IoT internship focusing on hardware-software integration using Texas Instruments Launchpads.

EDUCATION

MSc Advanced Computer Science with Industry

University of Leicester

Sep 2021 — Sep 2023

Passed with Merit. Course topics include Cloud Computing and Mobile and Web Applications.

BTech Electronics and Communications Engineering with Specialization in IoT and Sensors

VIT Vellore

Jul 2017 — Jul 2021

CGPA: 9.02/10. Course modules include DSA, Java, C++, Python, Web Technologies and Mobile App Development.

SKILLS

CI/CD & Automation: GitHub Actions, Azure DevOps Pipelines, Tekton

Containers & Orchestration: Docker, Docker Compose, Kubernetes, K3s, Helm

Cloud Platforms: Azure, AWS, GCP

Infrastructure as Code: Terraform, Ansible

Platform & Networking: Traefik, Nginx, Keycloak, OAuth2-Proxy, Slurm

Languages: Python, Go, JavaScript, TypeScript, Bash, HCL, Java

Frontend & Backend: React, FastAPI, Node.js, PostgreSQL

DevOps Tooling: Git, Linux, Nuitka, Zot (OCI Registry)

CERTIFICATIONS

- Kubernetes and Cloud Native Associate (KCNA)
- Architecting with Google Kubernetes Engine Specialization
- Architecting with Google Compute Engine Specialization
- Industrial IoT on Google Cloud Platform
- Java for Android